HEALTHCARE ANALYTICS MINI PROJECT (CA-III)

Predicting Dementia Using Brain Scan Measurements and Patient Health Data

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0.793
AUC with both kinds of data (imaging only 0.639, clinical only 0.742)

34% / 66%
Share of SHAP influence: imaging / clinical

0.938
AUC on OASIS-1 with the Random Forest

0.851
Best AUC on OASIS-2 (Logistic Regression)

THE PROBLEM

Dementia is usually spotted with a brain scan or a memory test, but the two are often looked at separately. If brain scan measurements and patient details go into one model, does it tell demented and nondemented people apart better, and which kind of information does it rely on?

THE DATA

OASIS-2 (training): 150 people aged 60 to 98, 373 visits.

OASIS-1 (second test set): 198 records, age 60+, with a CDR score.

Label: demented or nondemented ('converted' counted as demented; OASIS-1 CDR above 0).

Classes nearly balanced (51% / 49%). Missing SES (19) and MMSE (2)

TWO KINDS OF DATA, ONE MODEL

Brain scan

eTIV: size of the skull cavity
nWBV: how much of it the brain fills
ASF: scaling factor from eTIV

Patient details

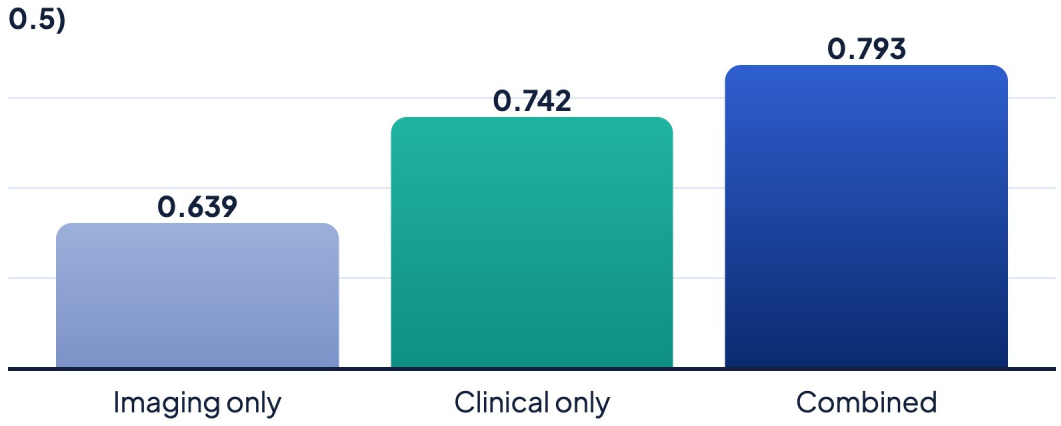
Age and **Sex**
SES: socioeconomic status
MMSE: 30-point memory test

HOW IT WAS DONE

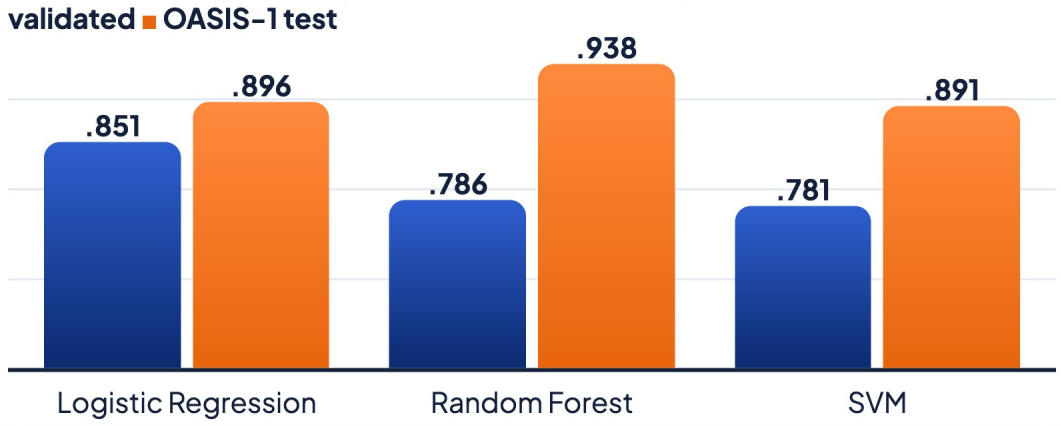
1. Explore and clean the data, check missing values and class balance.
2. Split by person: all visits of one person stay together.
3. Compare Logistic Regression, Random Forest and SVM (5-fold, 5 seeds).
4. Compare imaging only, clinical only and combined features.
5. Explain the Random Forest with SHAP.
6. Test on OASIS-1, a different group of people.
7. Deploy as a Streamlit dashboard.

RESULTS

Both kinds of data beat either alone (ROC-AUC, mean of 5 seeds, axis from 0.5)



Models on two datasets (ROC-AUC, axis from 0.6) ■ OASIS-2 cross-validated ■ OASIS-1 test





SHAP influence on average (imaging 31% to 37% across 5 splits). Low MMSE and low brain volume (nWBV) push towards demented.

WHAT IS NEW HERE

SHAP influence is added up separately for brain scan and clinical details. The closest paper we reviewed used SHAP on a mixed table without this split.

Splitting by person matters: a plain random split inflated Random Forest AUC to about 0.91, against about 0.79 when each person's visits stay together.

Tested on a second dataset, then deployed as a live dashboard with a SHAP chart per person.

WHAT THE RESULTS SAY

MMSE has the biggest influence in every split; it is closely tied to how the label is assigned.

Clinical details carry about two thirds of the influence, brain measurements one third.

On OASIS-2 the models miss about 3 in 10 demented cases (recall about 0.7).

Small dataset (150 people), age 60+. SHAP shows what the model used, not what causes dementia.

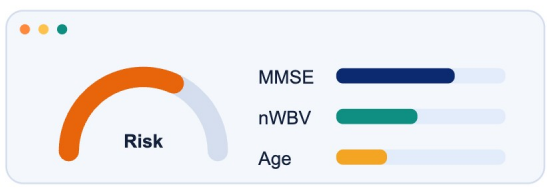
CONCLUSION

Combining brain scan and patient details predicted dementia status better than either alone (AUC 0.79 against 0.64 and 0.74). The Random Forest also held up on OASIS-1 (AUC 0.94). Most of the prediction comes from clinical details.

Future scope: more people and datasets, a threshold tuned to catch more cases, extra measurements, testing on follow-up visits.

LIVE DASHBOARD

Enter brain scan measurements and patient details to get a risk estimate and a chart of what pushed it.



App: dementia-risk-predictor-e5ozaf75khdibrrdpxkfqk.streamlit.app
Code: github.com/chirag-2025/dementia-risk-predictor

A learning project, not a medical tool. Mock-up shown for illustration.